

Irrigation technology, irrigation dose, and crop genetic impacts on alfalfa yield and quality

Bradley S. Crookston^a, Dakota Boren^a, Matt Yost^{a,*}, Tina Sullivan^b, Earl Creech^a,
Burdette Barker^c, Cheyenne Reid^d

^a Utah State University, Plants, Soils, and Climate Department, Logan, UT, USA

^b Kansas State University, Northeast Research Ext Center, Manhattan, KS, USA

^c Utah State University, Civil and Environmental Engineering Department, Logan, UT, USA

^d Utah State University, Extension Uinta Count, UT, USA

ARTICLE INFO

Handling Editor: Dr. Brent Clothier.

Keywords:

Alfalfa
Irrigation technologies
Deficit irrigation
Drought genetics
Water optimization

ABSTRACT

In water limited environments, alfalfa (*Medicago sativa*) is often criticized for its high water use, prompting interest in optimizing irrigation technologies, deficit irrigation, and drought-tolerant genetics. However, potential cumulative benefits from combining water-saving strategies have not been previously identified. This study evaluated the independent and combined effects of five irrigation technologies (low-elevation Nelson advantage, low-elevation precision application, low-elevation spray application, mid-elevation spray application, and mobile drip irrigation), four irrigation doses (growers' typical full dose, a 25 % reduction, and two 50 % reductions, uniform and growth stated-targeted), and two alfalfa varieties (growers' conventional and drought-tolerant) across three Utah sites from 2020 to 2022. No interaction effects were found among these factors, indicating that stacking multiple water-saving strategies did not enhance yield or forage quality. Low-elevation sprinkler technologies generally outperformed mid-elevation and mobile drip irrigation, though results varied by environment. Deficit irrigation at 25 % reduction often maintained yields similar to growers' Full irrigation dose, while 50 % reductions consistently decreased yield by 22–54 %. However, deficit irrigation improved forage quality and water use efficiency. Decision tree models revealed that maximizing relative feed value-adjusted water use efficiency primarily depended on matching irrigation dose and technology to site-specific climate demand rather than applying Full irrigation. These findings suggest that moderate deficit irrigation and low-elevation sprinkler technologies can improve forage quality and water resource efficiency without substantial yield loss that occurs with 50 % deficit irrigation.

1. Introduction

Agricultural production is often devastated by drought, causing yield losses of 30–90 % (Hussain et al., 2019). As drought continues across the western United States since 2004 and a mega-drought during the 2020–2022 growing seasons (Seager et al., 2022; Williams, Cook, and Smerdon, 2022), the effects have been expressed through irrigation curtailment, aquifer depletion, wells going dry, and reservoirs drying up. These compounding challenges raise the need for agricultural water management optimization and planning to maintain crop yield while water scarcity increase across the western United States and many other regions of the world.

Irrigation has been an integral part of global agriculture for

thousands of years. Global irrigated land grew two-fold from 1950 to 1970 because of the green revolution and a large increase in the number of dams (Sojka et al., 2002). The expansion in total irrigated land has slowed greatly since 1980. This has created a need to increase crop production effectively and efficiently on existing irrigated land to provide for a growing population. In the 20th century, farmers, engineers, and others began rapidly producing irrigation innovations. In a short span of 25 years (1935–1960), wheel lines, center pivots, and drip tape were invented (Waller and Yitayew, 2016). The use of center pivots and lateral-move systems are increasing due to their reduced labor requirements, generally good application efficiency and uniformity, and reliability. These mechanical-move systems play a key and growing role in the irrigation of field crops in the western United States and globally.

* Correspondence to: Utah State University, 4820 Old Main Hill, Logan, UT 84322, USA.

E-mail address: matt.yost@usu.edu (M. Yost).

<https://doi.org/10.1016/j.agwat.2025.109366>

Received 20 February 2024; Received in revised form 5 February 2025; Accepted 8 February 2025

Available online 26 February 2025

0378-3774/© 2025 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY-NC license (<http://creativecommons.org/licenses/by-nc/4.0/>).

Pivots have a wide variety of irrigation emission device technologies that vary in application efficiency and uniformity for different crop types, making them customizable to various cropping systems, soil types, and climates.

Most of the western United States is too arid to produce crops without irrigation. One of the main crops in Utah and other western states is alfalfa, which is often criticized for its water use. With > 75 % of diverted water resources being used for agriculture in many areas, increasing pressure exists to reduce agricultural water use (Dieter et al., 2017). Although alfalfa is a large water user, it has over \$418 million in sales annually in Utah, which is 13 times greater than the next highest-grossing crop for the state (USDA, 2021). It is also an important crop in sustaining animal agriculture. Total agricultural sales bring roughly \$21 billion to the Utah economy by the time products reach the table (Ward and Salisbury, 2016). The alfalfa market is well-developed, and storage of the harvested product is relatively simple. Utah also has a climate suited to growing high-quality alfalfa that is shipped throughout the U.S. and the world.

The western United States is not the only area looking for ways to optimize water use and improve crop irrigation. The Ogallala aquifer has been decreasing for many years in Texas, Oklahoma, Kansas, and Nebraska (Taghvaeian et al., 2017). Other areas along the Columbia River, Texas, and many areas in the U.S. Southwest Desert are experiencing declining groundwater levels. On a global scale, China is experiencing similar challenges with drought and is the location of many deficit irrigation trials (Li et al., 2017; Liu et al., 2021).

Many agriculture water optimization approaches have been investigated to help deal with water shortage and drought in the western United States and many other regions of the world. These include conversion to more efficient irrigation systems, improvements in irrigation management, alternative crops, deficit irrigation, fallowing, and other crop and soil management practices (Colby et al., 2014; Samani and Skaggs, 2006; Udall and Peterson, 2017; Yost et al., 2020). Three practices with the potential to optimize water use in alfalfa include crop genetics with greater drought tolerance, advanced sprinkler systems for center pivots and lateral-move systems, and deficit irrigation.

1.1. Genetics

Many alfalfa producers use current genetic advancements like pest and disease resistance, glyphosate resistance, low lignin, or multifoliate properties. Although a severe drought does not kill most varieties of established alfalfa, few can reach high yield potential under water restriction (Frate et al., 1991). It could be that many alfalfa breeders do not prioritize water efficiency or drought tolerance during the variety selection process. However, water use efficiency may become a more pressing need when developing new varieties. For example, New Mexico State University researchers are working on new alfalfa varieties adapted to the hot, dry New Mexico climate (KRWG News and Partners., 2014).

Most crops, including alfalfa, have been bred to maximize yield and quality given sufficient water and nutrient inputs (Dietz et al., 2021). However, this selection may lead to significantly reduced performance when inputs are below optimum levels. South Dakota State University has some *Medicago falcata* species that are very drought-tolerant but do not have desirable yields (Anower et al., 2017). They also found a wild variety of alfalfa on the Grand River National Grassland that has responded well to drought stress and had a much higher water efficiency (WUE) compared to 11 other accessions, two of which were commercialized. This could lead to beneficial cultivars for production because one of the concerns with previous breeding is that many of the drought alleles have been bred out of popular cultivars, and, thus, this genetic diversity has been lost (Dietz et al., 2021). One of the most common alfalfa varieties with enhanced drought tolerance is Ladak II. This variety has not been rigorously tested with other varieties in irrigation experiments to determine its ability to enhance drought tolerance under

varying water stress conditions.

1.2. Irrigation technology

Recent advancements in sprinkler technology provide another avenue for optimizing water use. There are four main categories of irrigation emission technologies for mechanical move systems. These categories are top-of-lateral, mid-elevation, low-elevation, and mobile drip irrigation (MDI). Top-of-lateral sprinklers are no longer common in the western U.S. because of wind drift and evaporation losses. Mid-elevation sprinklers include mid-elevation spray application (MESA), which is the most common sprinkler system used on pivots in Utah and much of the western United States. These sprinklers are installed at approximately 2–4 m spacing and vary from 1.0 to 2 m above the soil surface. These sprinklers have a wide wetted diameter ranging from about 6–20 m. This is often the least expensive type of system installed on a pivot, costing less than US\$6000 for a standard 400-m long pivot.

Low-energy precision application (LEPA) can have various spray patterns, including bubblers, accelerators, plates, and drag socks. Emission device spacing is roughly 1 m, and they are installed with the emission device positioned about 0.3–0.6 m above the soil surface. Because of the narrow spacing, it requires more sprinklers and other materials, increasing costs to around US\$9000–14,000.

Another low-elevation system is low-elevation spray application (LESA). It is a simple sprinkler design that incorporates a larger spray pattern of about 3–6 m, with the sprinklers positioned at 0.3–1 m above the soil surface. The spacing between sprinklers may be slightly wider than that of LEPA but is still typically less than 2 m. The sprinkler design and wider spacing reduce the cost slightly to US\$6000–8000 per pivot. One specific type of LESA device is the low-elevation Nelson Advantage (LENA) is a trademarked sprinkler by Nelson Irrigation, Inc. (Walla Walla, WA). The LENA system has a wider spacing of 2–3 m between sprinklers and is positioned at 0.6–1.0 m above the soil surface. The spray pattern is 9–18 m, and water is applied as rotating streams.

Mobile drip irrigation (MDI) is the most expensive system to purchase and install, ranging from \$20,000–30,000 per pivot. Drip tubing is pulled by a manifold fitted to the pivot or lateral-move system. Tubing length is adjusted to set irrigation rates, so the farther away from the center, longer tubes accommodate increasing rates. Emitters with 3.8–7.6 Lpm output are placed every 0.16 m on the tube, and the tubes are spaced about 1 m apart (Holt et al., 2021). It is necessary to have adequate filtration to prevent emitter clogging.

The application efficiency varies greatly among irrigation technologies. MESA has reported irrigation application efficiency of about 78 % due to more wind drift and evaporation than other systems due to its higher elevation sprinklers and wide spray patterns (Amosson et al., 2011). In the same study, LEPA application efficiency was measured at approximately 95 % because of the larger water stream and concentrated application pattern. Further, LESA efficiency was 88 % due to the wider spray pattern and smaller water droplets. LEPA and LESA are not as affected by wind as MESA sprinklers. Lastly, MDI was measured as the most efficient at 97 %. Direct irrigation application on the soil surface by MDI eliminates wind drift and minimizes evaporation before a crop canopy forms (Oker et al., 2020).

Several studies have compared the performance of various irrigation application systems. For example, researchers in Texas compared MDI to LESA and LEPA systems and found that in dry years, MDI had a higher WUE and was able to maintain sorghum [*Sorghum bicolor* (L.) Moench] grain yield (O'Shaughnessy and Colaizzi, 2017). While several research trials of these systems exist, they have not been rigorously tested together at various and replicated irrigation rates in alfalfa.

1.3. Deficit irrigation

Deficit irrigation is the application of less water than the evapotranspiration (ET) demand of a plant (Fereres & Soriano, 2006). Deficit

irrigation of alfalfa has been studied in the Western and Great Plains regions of the U.S. for many decades. In alfalfa systems with four cuttings or less, a large portion (40–55 %) of the annual yield is harvested in the first cutting. Add the second cutting, and it can represent up to 70 % of the annual yield. Most studies show that deficit irrigation causes yield loss, so if the loss can be avoided or reduced until the naturally lower-yielding cuttings, it could cause less yield loss than spreading insufficient water over the entire year (Montazar et al., 2020). Deficit irrigation of alfalfa could also be mitigated with more efficient sprinkler technologies. Many low-elevation sprinklers have reported efficiencies exceeding MESA by nearly 20 %. Thus, in theory, a 20–25 % reduction of water with low-elevation sprinklers should be full irrigation with MESA. Likewise, a 50 % reduction for low-elevation sprinklers over MESA could yield roughly the same as a 25 % reduction using MESA. We found no literature reporting the ability of low-elevation sprinklers or MDI to reduce irrigation water requirements in alfalfa. However, these technologies could provide substantial water savings for producers struggling with limited water supplies or who might be interested in participating in water banks or markets.

A report by Trout et al., (2020), included two primary purposes and strategies for deficit irrigation. First, the need to maximize net income from expensive water, and second, to irrigate more land with less water. Using surplus water from deficit irrigation on new land is often not an option for producers due to legal restrictions, but deficit irrigation because of expensive or unavailable water is already occurring. Another look at the practical application of deficit irrigation comes from a comprehensive study by Geerts and Raes (2009), where a list of deficit irrigation timings was compiled for many crops. They explain when it would be advisable to use deficit irrigation based on the growth stage of the plant. They also pinpointed some of the worst growth stages to stress crops. The only recommendation made for alfalfa is to alternate cuttings with full irrigation and deficit irrigation.

The western U.S. and most other regions with irrigated agriculture have limited studies focusing on drought-tolerant genetics, pivot irrigation technologies, and deficit irrigation for alfalfa, the most widely grown forage crop. Studies have investigated some of these factors in isolation, but none have combined all these potential water-saving techniques to determine how interactions affect alfalfa production and impact cumulative water savings. Thus, the objectives of this research were to evaluate how two alfalfa cultivars, five pivot irrigation technologies, and three deficit irrigation practices influence alfalfa yield and quality.

2. Materials and methods

2.1. Site characteristics and alfalfa establishment

Three sites were selected across the state of Utah for this study to represent various soil types and climates. Soil classification and other soil information from the NRCS soil survey were collected from the University of California-Davis SoilWeb application (O'Geen, 2020; Table 1). New alfalfa was seeded at all three locations at the beginning of the study. Two varieties were seeded at each site to represent a

drought-tolerant variety (Ladak II) and a non-drought-tolerant variety commonly recommended by crop advisors in each area where the sites were located. Ladak II was derived from Ladak and marketed as Ladak II by Dairyland Seed® (Kewaskum, WI). The Logan, UT site was established in 2019, and alfalfa was seeded in the fall of 2019 following the harvest of an oat (*Avena sativa* L.) forage crop. Soils at the Logan site had a pH of 7.4, organic matter of 27 g kg⁻¹, and salinity of 0.57 dS m⁻¹ at the beginning of the study. Two alfalfa varieties, Nexgrow 3605Q (conventional) and Ladak II (drought-tolerant) (Table 3), were each planted in three randomly selected strips measuring 9 × 239 m. Alfalfa seeding rates were 18.0 kg ha⁻¹ for both varieties, and both were planted with a John Deere drill (John Deere Manufacturing, Moline, IL) at a depth of 13 mm.

In 2020, a second site was established in Vernal, UT. This site had the highest organic matter of the sites, at 31 g kg⁻¹. The salinity was low at 0.58 dS m⁻¹, and the soil pH was 7.6. Two alfalfa varieties (Ladak II and WL 319 HQ) were seeded in the spring of 2021 at 22.4 kg ha⁻¹ using a John Deere drill (John Deere Manufacturing, Moline, IL) at 13 mm seeding depth. Both varieties were planted in three randomized strips measuring 9 × 183 m each. The third site was in Cedar City, UT and was established in 2021. Cedar City had the lowest organic matter and highest salinity of the sites, 18 g kg⁻¹ and 1.5 dS m⁻¹, respectively. Soil pH was like the other sites at 7.6. In the spring of 2021, two varieties (Ladak II and IFA 414) were sown using a Great Plains grain drill (Great Plains Ag, Salina, KS) at a rate of 20.1 kg ha⁻¹ and a depth of 13 mm. The randomly replicated strips measured 9 × 215 m each.

A single composite soil sample consisting of 18 subsamples representing the 0–0.30 m depth was collected each spring across the entire trial area at each site. The macronutrient and micronutrient concentrations were measured, and fertilizer was applied when needed according to Utah State University Extension fertilizer guidelines (Cardon et al., 2008). Herbicides were applied once or twice a year, depending on weed species and weed pressure at each site (Table 2). Most insect damage was minimal in all site years other than at Logan in 2022. That year, there was noticeable damage just before the first cutting, and it was decided it would be more cost effective to harvest a few days early

Table 2

Management practices utilized at three sites in Utah during 2020–2022. Herbicides were applied according to label instructions. The fertilizer was applied based on soil test results per Utah State University Extension guidelines.

Site	Year	Fertilizer products and rates ^a	Herbicide product(s)
Logan	2020	None	Raptor, Class Act, Superb
	2021	112 kg ha ⁻¹ MAP, 151 kg ha ⁻¹ Potash	Raptor, Class Act, Superb
	2022	None	None
Cedar City	2021	112 kg ha ⁻¹ MAP, 22 kg ha ⁻¹ Potash	Glory, Killzall
	2022	None	None
Vernal	2021	None	None
	2022	None	Rodeo

^a Monoammonium phosphate (MAP)

Table 1

Site and soil properties for the three study locations in Utah from 2019 to 2021.

Nearest town	Years	Coordinates	Elev.	Dominant soil texture (classification)	Soil OM ^a	Soil pH	Salinity
Logan	2020–2022	41.667, -111.920	m 1365	Loam (Fine-silty, mixed, mesic Aquic Calciustolls)	g kg ⁻¹ 27	7.4	dS m ⁻¹ 0.57
Vernal	2020–2021	40.462, -109.562	1667	Clay loam, loam (Fine-loamy, mixed, superactive, calcareous, mesic Typic Torriorthents)	31	7.6	0.58
Cedar City	2021–2022	37.665, -113.142	1682	Sandy loam, silty clay loam (Coarse-loamy, mixed (calcareous), mesic Xeric Torriorthents)	18	7.6	1.6

^a OM, organic matter

rather than spray insecticide. None of the varieties were glyphosate resistant, and all the varieties had the same fall dormancy rating as Ladak II (fall dormancy 3).

Site daily precipitation, air temperature, and reference evapotranspiration were obtained from the Utah Climate Center (Agweather, 2022). Weather data were used to calculate cumulative precipitation, average monthly temperature, and estimated evapotranspiration during the alfalfa irrigation season (1 April – 30 September) for the 2020 – 2022 study period. Conditions were compared to the National Oceanic and Atmospheric Administration (Silver Springs, MD) 30-year normal (1991 – 2020) for precipitation and air temperature.

2.2. Irrigation technologies

The experimental design was a nested factorial with random factors. Irrigation rate was nested within irrigation technology as there was limited area in a four-to-five-span lateral-move irrigation system and randomization of various technologies was not possible due to irrigation technology wetting patterns. All irrigation was conducted with Zimatic laterals (Lindsay Corporation, Omaha, NE). The irrigation technologies were randomly placed on individual spans. The included sprinkler types varied slightly by site-year due to regional interest and stakeholder input.

In Vernal, the four-span lateral had LEPA, LESA, MESA, and MDI in 2020 and 2021. For the 2022 growing season, the MDI system was replaced with LENA due to logistical constraints in maintaining proper irrigation rates with MDI despite continuous efforts to filter water with automated filters with fine (150-micron) screens. The Cedar City site had a four-tower lateral with LENA, LESA, LEPA, and MESA. The Logan site have five towers with LEPA, LESA, LENA, MESA, and MDI randomly placed on each span.

2.3. Irrigation dose treatments

Irrigation treatments were imposed by using different nozzle sizes within each lateral span. Every lateral span was partitioned into four randomly located sections that were approximately 12 m long corresponding to the four different irrigation doses. The Full irrigation dose was based on the experience of farm managers and allowed water turns at the site. The three deficit doses were intended to be 75 % of the full dose, and two doses at 50 % of full irrigation, namely, Moderate, Limited U (uniform) and Limited T (targeted, Table 3). The Moderate and Limited U treatments were a uniform deficit for the entire irrigation season, while the Limited T dose was targeted irrigation that applied differing amounts of water at different times of the year. The desired design of targeted deficit was for a full dose to be applied during the first half of the season, then no irrigation after the second cutting to create an average of 50 % applied seasonal water. The goal with this treatment was to take advantage of the greater water use efficiency in the first half of the season (Ottman and Putnam, 2017). Due to water uncertainty and water turns, the desired 50 % targeted reduction was not always 50 % but ranged from a 22–51 % total reduction across site-years (Table 3). Further, the practice of ceasing irrigation in the second half of the season was only implemented in the first and second full production years and not the establishment year to avoid damaging alfalfa seedlings and young plants and to ensure adequate irrigation during establishment. In addition, alfalfa was irrigated at 100 % in all treatments during the first one to three irrigations after seeding to ensure even germination and establishment before irrigation stress occurred to avoid damaging or killing plants. Total water was the sum of growing season precipitation and irrigation applied (TW, Table 3). Water use efficiency (WUE) was calculated as dry matter yield divided by TW. While measured soil water is an important part of accurately estimating WUE, a soil water analysis was excluded for brevity because those data will be considered in a separate publication.

Table 3
Summary of total water inputs (precipitation and irrigation) and proportions of estimated reference crop evapotranspiration (ET_{rc} , m^3) met by irrigation or total water under four irrigation treatments (Full, Moderate, Limited U [uniform], Limited T [targeted]) at three Utah sites.

	2020					2021					2022				
	Full	Moderate	Limited U	Limited T	ET _o	Full	Moderate	Limited U	Limited T	ET _o	Full	Moderate	Limited U	Limited T	ET _o
Logan															
TW (m ³)	1838	1470	1103	1093	3033	1804	1431	1059	1136	2843	1622	1371	1121	1108	2830
% Irr	80	75	67	66		83	78	70	72		62	55	45	44	
% IrrET _o	48	36	24	24		52	39	26	29		35	27	18	17	
% TW ET _o	61	48	36	36		63	50	37	40		57	48	40	39	
Cedar City															
TW						2769	2241	1712	1845	4192	3859	3097	2335	2829	4162
% Irr						76	71	62	65		79	74	65	71	
% IrrET _o						50	38	25	28		73	55	37	48	
% TW ET _o						66	53	41	44		93	74	56	68	
Vernal															
TW						1326	1055	784	1093	2511	2572	2001	1429	1449	3605
% Irr						82	61	41	64		89	86	80	80	
% IrrET _o						43	32	22	34		63	48	32	32	
% TW ET _o						53	42	31	44		71	55	40	40	

Note, the Moderate and Limited U (uniform) irrigation doses were uniform reductions in irrigation all season. Whereas the Limited T (targeted) reduction was adjusted at full, half, or no irrigation to target a 50 % reduction at various growth stages.

TW, total water from irrigation and precipitation; % Irr, proportion of total water from irrigation; % Irr ET_o , proportion of ET_o from irrigation; % TW ET_o , proportion of ET_o from total water.

2.4. Alfalfa variety treatments

All plots at all sites were approximately 9×12 m, varying slightly from site to site because the span length of every lateral was not equal. Plots were positioned lengthwise across the field perpendicular to the travel direction of the lateral to create strip plots. There was a total of six strips in each field with three replications of two alfalfa varieties (Fig. 1). Alfalfa cutting schedules were determined by farm managers at each location and most were cut between the bud to first flower alfalfa growth stage to maximize yield and quality.

2.5. Harvest procedures

Harvest procedures varied among site-years based on equipment availability and other logistics. In 2020 and 2021 at the Logan and Vernal sites, whole strips were harvested with a field-scale mower. The plot strips were marked out to ensure the mower took a full cut down each strip. Once it was cut, a 3-m section of the windrow inside each plot was cut and weighed to determine bulk wet weights. This represented a harvest area of 15.5 m^2 . An alfalfa subsample (about 450 g) was collected from each plot. Subsamples were used to determine moisture, calculate dry matter yield, and conduct forage quality analysis. This was an effective but logistically difficult harvest method. To improve efficiency and reduce labor requirements, a walk behind Swift flail harvester (Swift Current Saskatchewan, Canada) was used in 2021 at Cedar City and at all sites in 2022. The Swift harvester had a 0.76-m cutting width, so two swaths 3 m long each were cut for a total harvest area of 4.7 m^2 in each plot. The subsample collection procedure remained the same throughout the study.

2.6. Forage quality calculations

Alfalfa subsamples were dried at 60°C until constant mass, weighed, and then ground to pass through a 1-mm sieve using a Thomas Model 4 Wiley Mill (Thomas Scientific, Swedesboro, NJ). Ground samples were analyzed by near-infrared reflectance spectroscopy (NIRS) using a FOSS DS2500 F (Foss North America Inc., Eden Prairie, MN) at the Utah State University Analytical Laboratories (Logan, UT), utilizing the 2020 legume hay NIRS consortium equations (NIRS Forage and Feed Consortium, Berea, KY) to estimate dry matter, ash, fat, crude protein (CP), acid detergent fiber (ADF), neutral detergent fiber (NDF), and neutral detergent fiber digestibility 48 hr (NDFD48). Total digestible nutrients

(TDN), relative feed value (RFV), and relative feed quality (RFQ) were then computed using equations 1, 2, and 3, respectively.

[Eq. 1]

$$TDN = [100 - (NDF - 2 + CP + 2.5 + Ash)] \times 0.98 + CP \times 0.93 + (Fat - 1) \times 0.97 \times 2.25 + (NDF - 2) \times \frac{NDFD}{100} - 7$$

[Eq. 2]

$$RFV = \frac{\left(\frac{120}{NDF}\right) \times (88.9 - 0.779 \times ADF)}{1.29}$$

[Eq. 3]

$$RFQ = \frac{\left[\left(\frac{0.012 \times 1350}{NDFD}\right) + \left(\frac{(NDFD - 45) \times 0.374}{1350}\right) \times 100\right] \times TDN}{1.23}$$

2.7. Description of statistical analyses

All statistical analyses were performed by site at $P \leq 0.05$ using the MIXED procedure of SAS (SAS Version 9.4, SAS Institute Incorporated, Cary, NC). Each site was analyzed separately because the treatments varied slightly among sites. In all analyses, site, year, irrigation technology, irrigation dose, genetics and most of their interactions were considered fixed factors. Block and interactions with blocks were considered random effects as was the irrigation technology $\times$ irrigation dose interaction because this interaction was restricted in its randomization due to restraints in the experimental design. Residuals were evaluated for homogeneity of variance and normality using scatterplots of residuals versus predicted values with the UNIVARIATE procedure of SAS. All mean separations were conducted using Fisher's protected LSD at $P = 0.05$ using the PDIF option of the MIXED procedure of SAS.

Since the ANOVA model could not formally test technology $\times$ irrigation dose as a fixed interaction, a bootstrapped regression trees machine learning approach, using the *rpart* package in R (Therneau et al., 2015), was used to evaluate key predictor thresholds and estimate a quality and efficiency outcome metric. Decision trees are inherently non-parametric and can uncover practical interactions (data splits) even when the experimental design does not allow for a formal statistical test of those interactions (Breiman, 2017). Furthermore, it is important to

	LESA				LEPA				MESA				MDI or LENA			
DT																
CONV																
CONV																
DT																
DT																
CONV																
	100%	T50%	75%	50%	T50%	50%	75%	100%	75%	T50%	100%	50%	50%	100%	T50%	75%

Fig. 1. Model of the plot layout for irrigation technology, irrigation rate, and alfalfa genetics treatments at three sites in Utah. Technology treatments included, low-elevation Nelson advantage (LENA), low elevation precision application (LEPA), low elevation spray application (LESA), mid-elevation spray application (MESA), and mobile drip irrigation (MDI). Rates included, full (100 %) and three deficit irrigation treatments at 75 % 50 % uniform and 50 % targeted reductions (T50 %). Alfalfa genetics included conventional (CONV), and drought-tolerant (DT).

note that while decision trees can reveal data patterns, the restricted experimental randomization between the technology and dose factors requires caution when interpreting causality. Still, the tree models provide valuable insights into how irrigation technology and various dosing patterns might work together to affect outcomes in the real-world data observed from this study. To model conditions that would lead toward optimal alfalfa production that is resource efficient and sustainable, the dependent variable was relative feed value-adjusted water use efficiency ($\text{RFV} \times \text{WUE Mg ha}^{-1} \text{ m}^{-3}$, WUE_{RFV}).

Data were prepared by first generating 100 bootstrapped subsamples from each site by randomly sampling with replacement. Bootstrapping was used to reduce overfitting and increase split reliability, enabling more robust inference (Tibshirani and Efron, 1993). Within each bootstrapped site-level subset, a regression tree was fit with alfalfa genetics, irrigation technology, and the proportion of ET_0 met by irrigation (IrrET_0 , %) as predictor variables. Predictors and conditions selected by the machine learning algorithm minimize the variance, thus the variable chosen at a split tends to explain more variability and is impactful when predicting the dependent variable (Breiman, 2017). The decision tree split information from all 100 models per site were subsequently summarized and pooled to identify the most frequent branching variables across bootstraps, thereby illustrating stable partition points within site-level data. After identifying common splits, a consensus decision tree was fit for each site. The final models were visualized using the *rpart.plot* package in R (Milborrow, 2018). The decision tree models are read from top to bottom with the left or right split

(yes/no) indicating the condition for the predictor variable. Nodes contain the predicted average WUE_{RFV} and the proportion of observations in that node.

3. Results and discussion

3.1. Site weather

Precipitation levels varied among all site-years, and all deviated from their 30-yr (1991–2020) normal precipitation. In Logan in 2020 and 2021, the growing season (1 April to 30 September) cumulative precipitation was 40 % and 30 % less than the 30-yr normal, respectively (Fig. 2). From May to August during 2022 at Logan, the cumulative precipitation was about 33 % lower than the 30-yr normal, however, by the end of the season, they were similar. The average daily average air temperature in Logan 2021 were 5°C warmer in June and 3°C in July than normal (Fig. 2). Logan, in 2022, started the growing season cooler than the normal through mid-May before quickly exceeding the normal and staying about 2°C hotter through the rest of the growing season.

Cedar City and Vernal both were below the 30-year normal precipitation during the early season. However, rainfall during late July through August increased precipitation totals at the end of the season beyond normal by 8 % at Cedar City in 2021, and 27 % in 2022. Likewise, at Vernal, precipitation was above normal by 12 % and 27 % in 2021 and 2022, respectively. Air temperatures in Cedar City during 2021 and 2022 were similar to the 30-yr normal until July and August

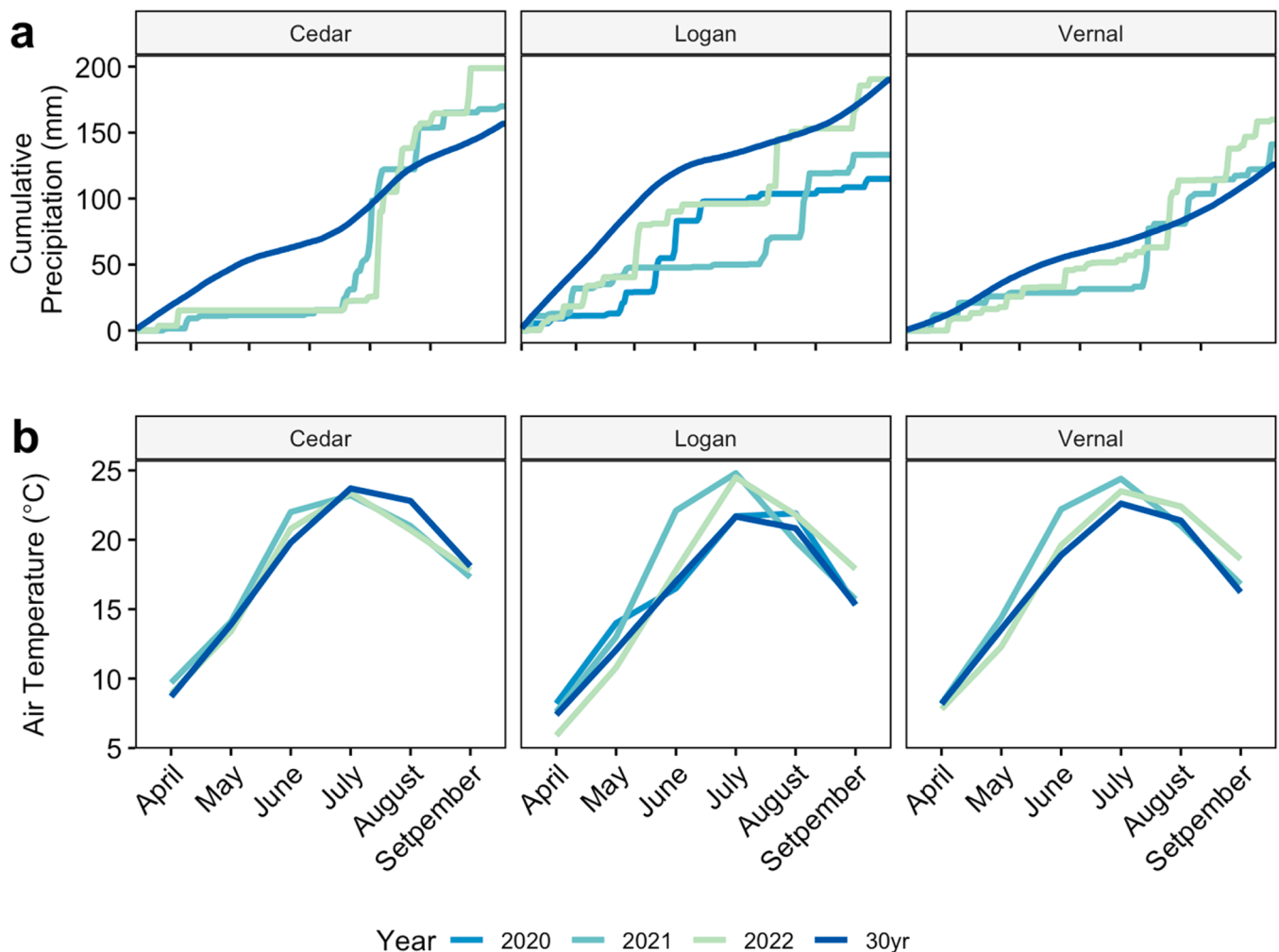


Fig. 2. Weather conditions at three sites in Utah during 2020–2022 for the alfalfa irrigation season (1 April - 30 September). This includes a) the daily cumulative precipitation (mm) and b) monthly average temperature (°C), along with their respective 30-year (1991–2020) normal.

when it was cooler than normal, coinciding with the late-season rains. Vernal in 2021 was a warmer than normal for most of the growing season with the peak in July of 2°C warmer than normal. In 2022 at Vernal, the daily average air temperature was cooler than normal early in the season and then warmer later in the season. Climatic demand estimated from the reference crop evapotranspiration (ET_o , Table 3) varied slightly by year at each site but was more consistent at Logan and Cedar City from 2021 to 2022. At Logan, ET_o was higher in 2020 compared to 2021 and 2022 by approximately 7 %. The ET_o at Cedar City varied by less than 1 %, however, at Vernal, climatic demand was 44 % greater in 2022 than in 2021.

3.2. Summary of interaction effects

A total of seven site-years of data were collected from three sites in Utah during 2020–2022. Data were collected from the establishment year and the first one, two, or three production years of alfalfa and should be considered in that context. One of the major objectives of this study was to determine if there were cumulative or ‘stacked’ benefits of combining drought-tolerant genetics and irrigation technologies for alfalfa production. In all seven site-years there were no two-way interactions between alfalfa genetics and irrigation technology (Table 4). This indicates that there were no cumulative benefits of these technologies for optimizing water use or improving alfalfa production at these site-years. The two-way interaction between irrigation technology and irrigation dose or the three-way interaction between these two approaches and alfalfa genetics could not be evaluated because of limitations in the experimental design that made it difficult to randomly place irrigation doses and irrigation technologies when working with field-

scale irrigation systems. Future research should incorporate replicated interactions of irrigation technologies and doses to clarify whether low-elevation sprinklers and MDI can reduce water needs while maintaining yield.

Another key objective was to determine whether the drought-tolerant variety (Ladak II) could enhance production under water-deficit conditions. Across all seven site-years, the genetics $\times$ irrigation dose interaction was also not significant (Table 4), indicating no measurable advantage of the drought-tolerant variety over the conventional varieties in terms of yield or forage quality. These findings align with prior research on deficit-irrigated alfalfa using a subsurface drip system, where differences in yield among 15 alfalfa varieties were found, but no genetic $\times$ irrigation interaction was detected (Radawich, 2018). While some studies have identified specific alfalfa varieties that exhibit superior drought resilience (Kang et al., 2011; Anower et al., 2017), the lack of interaction results from this study suggest that Ladak II does not confer meaningful drought tolerance benefits.

3.3. Alfalfa genetics effects

The main effect of alfalfa genetics influenced yield in four of the seven site-years (Table 4; Fig. 3). Yields varied by site-year, with the lowest observed at Cedar City in 2021 for the drought-tolerant variety (3.6 Mg ha^{-1}) and the highest at Vernal in 2022 (10.1 Mg ha^{-1}). During establishment years, the drought-tolerant Ladak II yielded 6.8 % and 10 % less than the conventional varieties at Cedar City in 2021 and Logan in 2020, respectively. However, in subsequent production years, Ladak II outperformed the conventional varieties by 8 % and 11 % at Cedar City and Vernal in 2022, respectively. Notably, there were no

Table 4

Significance of *F* tests for the fixed effects of irrigation technology, irrigation dose, alfalfa genetics, and their interactions on yield and crop quality parameters. Differences were considered statistically significant when $P < 0.05$.

Site	Year	Effect	Yield	WUE	CP	ADF	NDF	TDN	RFV
$P > F$									
Logan	2020	Technology (T)	< 0.001	< .000	0.077	< 0.001	< 0.001	< 0.001	< 0.001
		Dose (D)	< 0.001	0.004	0.027	< 0.001	< 0.001	< 0.001	< 0.001
		Genetics (G)	< 0.001	0.000	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
		T $\times$ D	0.921	0.955	0.326	0.377	0.343	0.377	0.96
		D $\times$ G	0.29	0.269	0.567	0.383	0.368	0.381	0.572
Logan	2021	T	< 0.001	< .000	0.115	< 0.001	< 0.001	< 0.001	< 0.001
		D	< 0.001	< .000	0.281	0.011	0.042	0.01	0.033
		G	0.086	0.085	< .000	< .000	< 0.001	< 0.001	< 0.001
		T $\times$ D	0.681	0.770	0.606	0.477	0.641	0.476	0.744
		D $\times$ G	0.822	0.759	0.078	0.322	0.266	0.321	0.532
Logan	2022	T	< 0.001	< .000	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
		D	0.09	< .000	0.134	0.015	0.031	0.007	0.019
		G	0.159	0.128	0.001	0.002	0.01	< 0.001	< 0.001
		T $\times$ D	0.977	0.992	0.599	0.721	0.841	0.649	0.641
		D $\times$ G	0.345	0.237	0.871	0.887	0.911	0.735	0.889
Vernal	2021	T	0.002	0.003	0.007	0.1	0.001	0.115	0.004
		D	< 0.001	< .000	0.387	< 0.001	0.013	< 0.001	0.004
		G	0.358	0.355	0.535	0.037	0.407	0.027	0.969
		T $\times$ D	0.946	0.917	0.217	0.352	0.235	0.418	0.165
		D $\times$ G	0.485	0.512	0.513	0.883	0.595	0.758	0.723
Vernal	2022	T	0.029	0.027	0.042	0.015	0.02	0.035	0.014
		D	0.014	< .000	0.003	0.002	0.003	0.042	0.004
		G	0.001	0.002	0.001	0.015	0.027	0.01	0.018
		T $\times$ D	0.837	0.804	0.082	0.653	0.562	0.222	0.448
		D $\times$ G	0.683	0.823	0.207	0.066	0.071	0.076	0.131
Cedar City	2021	T	0.75	0.765	0.029	0.149	0.689	0.149	0.661
		D	< 0.001	0.116	0.679	0.003	0.004	0.003	< 0.001
		G	0.019	0.021	0.034	0.037	0.003	0.037	0.003
		T $\times$ D	0.522	0.417	0.279	0.354	0.182	0.354	0.369
		D $\times$ G	0.937	0.937	0.097	0.713	0.417	0.713	0.35
Cedar City	2022	T	0.194	0.217	0.034	0.212	0.229	0.274	0.134
		D	0.001	0.060	0.001	0.006	0.02	0.019	0.005
		G	0.031	0.025	0.001	0.001	0.001	0.001	0.001
		T $\times$ D	0.422	0.381	0.289	0.455	0.371	0.421	0.3
		D $\times$ G	0.717	0.589	0.909	0.969	0.992	0.912	0.88

Note: WUE, water use efficiency; CP, crude protein; ADF, acid detergent fiber; NDF, neutral detergent fiber; TDN, total digestible nutrients; RFV, relative feed value.

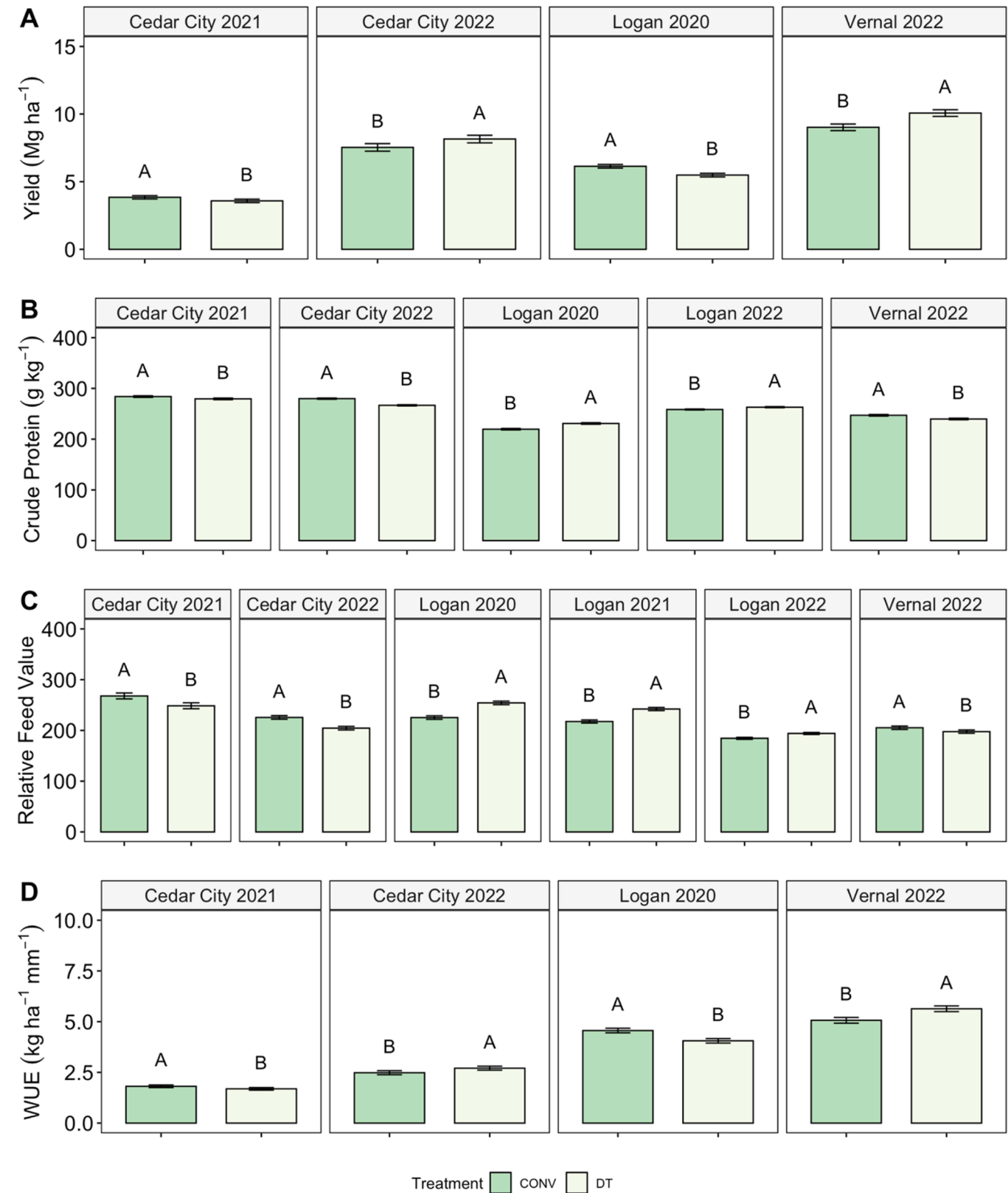


Fig. 3. The effect of conventional (CONV) and drought-tolerant (DT) genetics on alfalfa dry matter yield (A), crude protein (B), relative feed value (C), and water use efficiency (WUE; D). Error bars represent standard errors. Letters above bars represent mean separations conducted by site-year.

differences in yield between varieties at Logan in years following establishment. These results suggest Ladak II may prioritize root growth during establishment, potentially at the expense of vegetative production. Evidence for this hypothesis was shown by researchers at South Dakota State University where drought-tolerant varieties experienced smaller reductions in root mass under water stress compared to other varieties (Anower et al., 2017).

For forage quality, the drought-tolerant and conventional varieties frequently achieved or exceeded the threshold for 'supreme' quality alfalfa (220 g CP kg⁻¹; USDA, 2023). Genetic effects influenced crude protein (CP), relative feed value (RFV), and several other quality parameters in six of seven of site years, with Vernal 2021 as the only exception. In Logan 2020, CP for the drought-tolerant variety was 5.5 % higher than the conventional. The elevated CP in this case could increase alfalfa market value, however, the it may not translate into overall economic value due to the slight difference. For most site-years, genetics impacts on CP were modest, with larger variations observed in Logan in 2020 and 2021, where CP increased by 16 % and 14 %, respectively for the conventional and drought-tolerant varieties. The RFV trends mirrored those of CP (Table 4). At Logan (2020–2022), Ladak II

consistently outperformed the conventional variety with RFV improvements of 11 %, 10 %, and 5 % in consecutive years. Conversely, at Vernal in 2022 and Cedar City (2021, 2022), RFV for Ladak II was 4 %, 7 %, and 9 % lower than that of conventional varieties. Despite these differences, nearly all RFV values exceeded the threshold of 185 for 'supreme' quality alfalfa.

Ladak II exhibited slightly higher WUE during production years at Cedar City and Vernal in 2022. However, no differences were observed at Logan following the establishment year (2020). These results may suggest that the drought-tolerant variety advantages in improving WUE are dependent on environmental conditions. Specifically, the estimated climatic demand in 2022 was greater in Cedar City and Vernal compared to Logan by 47 % and 27 %, respectively. Additionally, the total water (TW) applied in Logan during the production years (2021 and 2022) was less than 48 % of climatic demand when averaged across irrigation regimes. Yet, in Cedar City and Vernal in 2022, TW and ET_o averaged 73 % and 52 %, respectively. It is likely that establishment-year root development conferred slightly greater WUE in the drought-tolerant variety during the higher stress environmental conditions. This suggests that the potential benefits of drought-tolerant genetics may be site-specific and

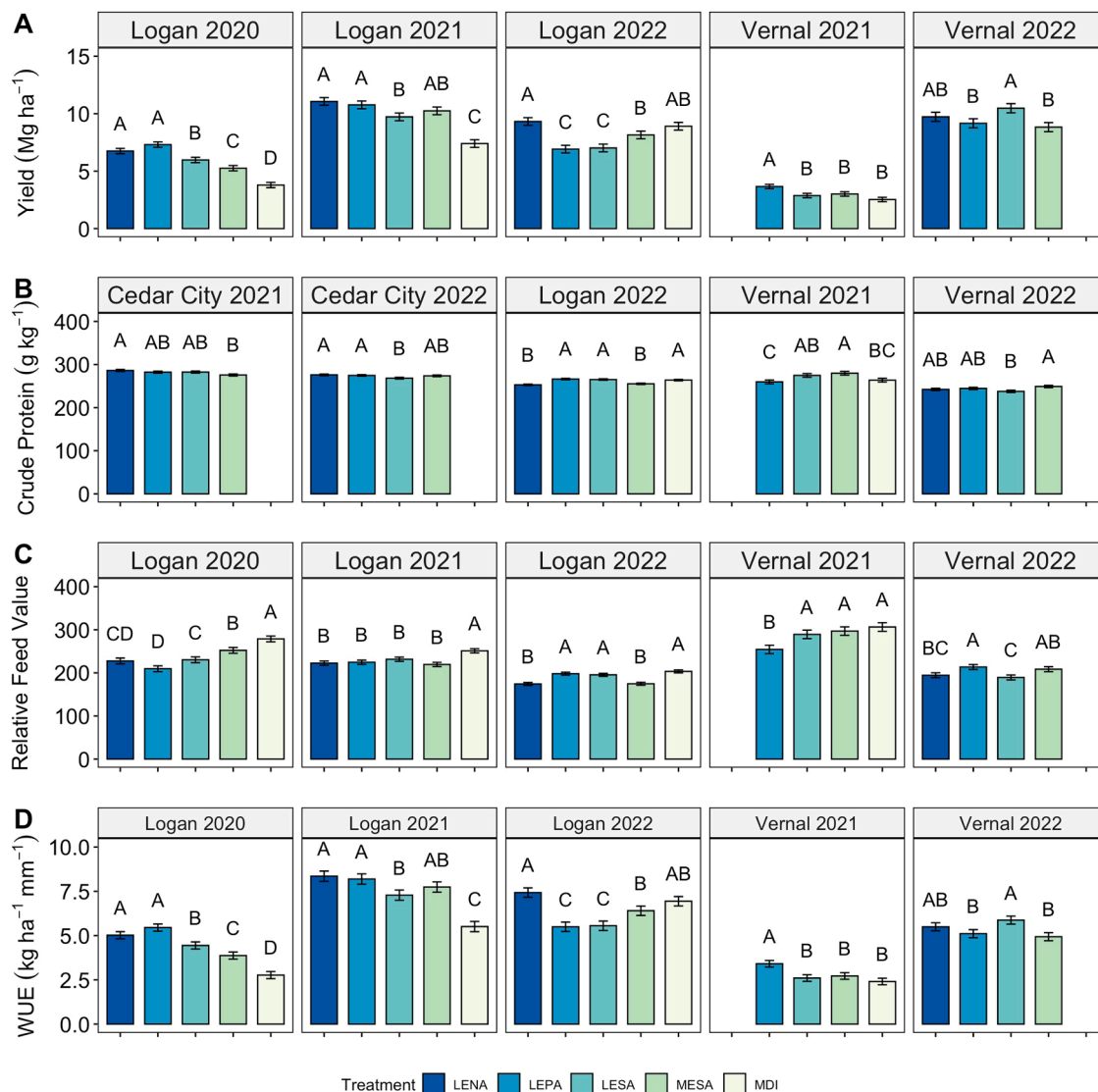


Fig. 4. The effect of irrigation technologies on alfalfa dry matter yield (A), crude protein (B), relative feed value (C), and water use efficiency (WUE; D). Technology treatments included, low-elevation Nelson advantage (LENA), low elevation precision application (LEPA), low elevation spray application (LESA), mid-elevation spray application (MESA), and mobile drip irrigation (MDI). Error bars represent standard errors. Letters above bars represent mean separations conducted by site-year.

dependent on environmental conditions rather than a generalizable trait across locations and years.

3.4. Irrigation technology effects

Irrigation technology influenced alfalfa yield in most site-years, with responses varying across environments (Table 4; Fig. 4). However, there was a pattern of low-elevation systems (LENA, LESA, and LEPA) frequently producing higher yields compared to MESA and MDI. At Logan and Vernal during the establishment years of 2020 and 2021, LEPA outperformed MESA by 14 % and 22 %, respectively. However, yield advantages diminished in production years, suggesting that irrigation technology benefits may be most pronounced in the establishment years. The lack of yield response to irrigation technology at Cedar City in 2021 was likely influenced by the abnormally high precipitation during the late summer (Fig. 2). This did not occur in 2022 and thus it was not clear why the technologies did not influence production that year. Water use efficiency perfectly mirrored the yield response to irrigation technology in magnitude and quantity of differences among treatments and site-years. These results demonstrated the potential for low-elevation systems (LENA, LEPA, and LESA) to improve yield and

WUE, particularly during production years. However, the benefits of these technologies were not universal across site-years, indicating that site-specific factors such as precipitation and soil properties influence the relative effectiveness of irrigation methods. MDI, while theoretically the most efficient, exhibited challenges that limited its performance (see Methods), emphasizing the importance of proper system management and maintenance.

The irrigation technologies had frequent but variable effects on all forage quality parameters in most cases at Logan and Vernal but not at Cedar City (ADF, NDF, TDN and RFQ; Table 4, data not shown). The technologies influenced CP in five site-years, albeit with minimal differences across treatments and failed to outperform one another in consistently producing higher CP levels (Fig. 4B). For example, LENA and LEPA had the highest CP values among treatments at Cedar City, and in Logan LENA had the lowest CP values. In the establishment year at Vernal (2021), the technology treatment differences were more distinct than at the other site-years with 7.7 % more CP from MESA compared to LEPA. However, alfalfa RFV had a reverse pattern at Vernal with more dynamic treatment responses observed in the production year versus the establishment year. Overall, RFV responded in the same site-years in Logan and Vernal as observed from alfalfa yield, however RFV

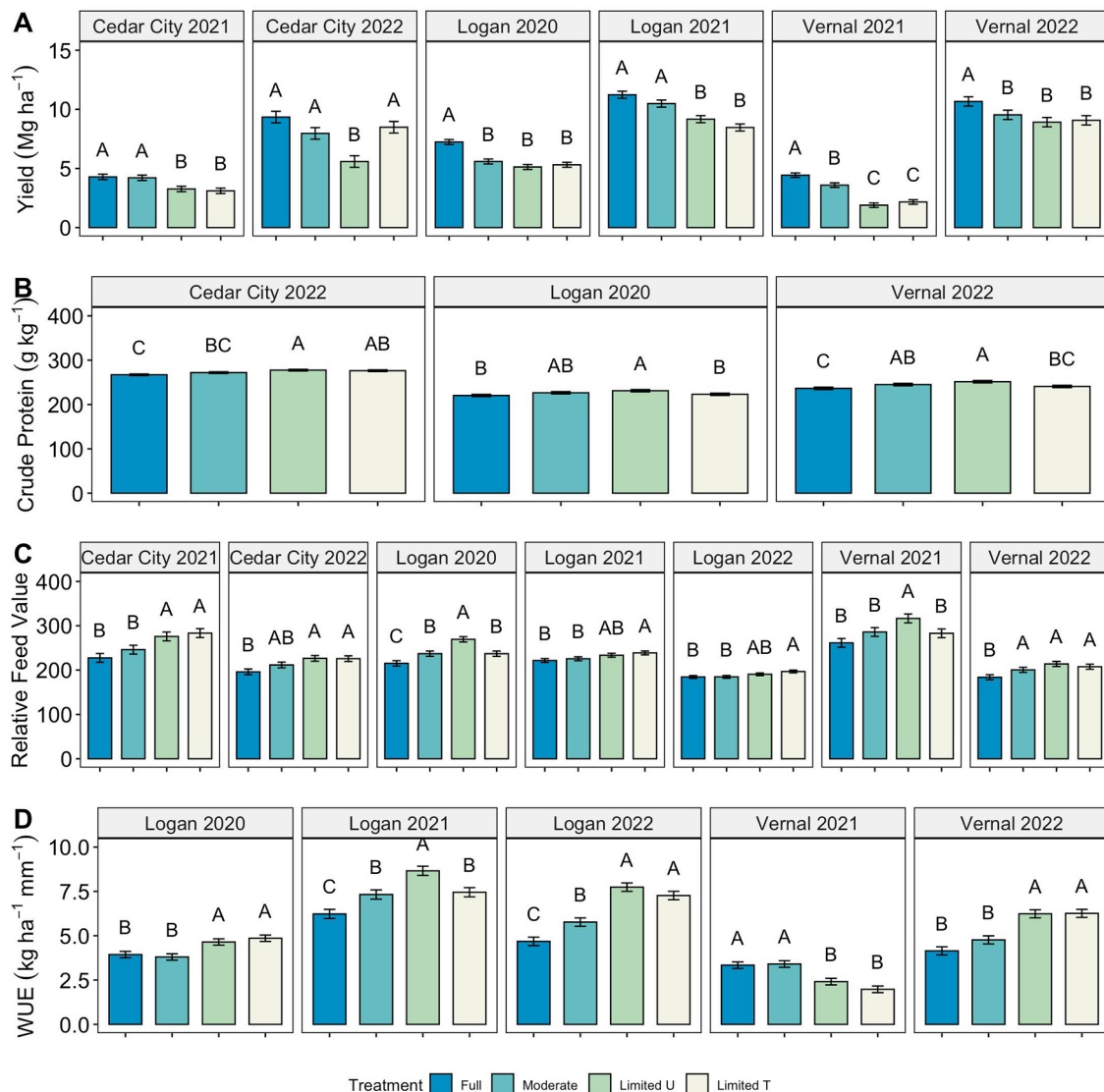


Fig. 5. The effect of irrigation dose on alfalfa dry matter yield (A), crude protein (B), relative feed value (C) and water use efficiency (WUE; D). Amounts included, High and three deficit irrigation doses at Moderate, Limited U (uniform), and Limited T (targeted) reductions. Error bars represent standard error. Letters above bars represent mean separations conducted by site-year.

had an inverse relationship to yield, confirming earlier findings that alfalfa quality is primarily a physiological consequence of water abundance (Ottman and Putnam, 2017) rather than a response to the irrigation delivery technology. It is likely that future soil water content analyses from this field experiment will provide additional insights on possible water distribution differences through the root zone under these technology treatments.

3.5. Deficit irrigation effects

Deficit irrigation influenced alfalfa yield in six site-years, with the Full and Moderate irrigation doses producing similar alfalfa yields in half of the significant seasons, however, there was a pattern of higher yield in production years compared to establishment seasons (Fig. 5). Yields in the 50 % reduction treatments (Limited U and Limited T) were similar across site-years except for 2022 at Cedar City. In that year, the proportion of total water as irrigation was 13 % less in the Limited U than the Moderate treatment due to early season irrigation. The largest average yield loss of 54 % was observed at Vernal 2021 in the Limited U and Limited T doses, while 56 % more irrigation was applied in the Full dose (Table 5). However, in 2022 at Vernal, average yield losses in the three deficit treatments were only 14 % less than the Full dose, yet irrigation averaged 71 % more under Full irrigation. Further, the Limited U at Cedar City in 2022, experienced an average of 45 % yield loss compared to the Full, Moderate, and Limited T doses with 48 % more water. Sill, across other site-years, the yield losses averaged approximately 22 %, while the irrigation averaged between 56 % and 66 % more than the lowest yielding deficit doses. These results agreed with previous research on alfalfa showing that there is almost always a reduction in yield due to decreased water input yet, the size of loss varies according to site environment (Orloff et al., 2005; Hanson et al., 2007; Lindenmayer et al., 2011), and is nonlinearly related to water abundance (Sitterson et al., 2023). Undoubtedly, a future analysis of soil water will show how irrigation doses affected increased water availability throughout the soil profile.

The WUE results further demonstrated diminishing returns of alfalfa water resource efficiency due to irrigation applied beyond an optimum dose (Sitterson et al., 2023; Fig. 5). While Full irrigation maximized absolute yield, WUE was often highest under the Limited regimes at Logan and at Vernal in 2022. Water use efficiency means were often grouped in Full and Moderate vs. Limited U and Limited T. Exceptions to the grouping pattern was observed in 2021 and 2022 at Logan where the Moderate had intermediate WUE but yield similar to the Full dose. Additionally, Vernal in 2021 was the only site-year that reversed the relationship of lower efficiency at higher irrigation doses, likely due to a combination of lower climatic demand and lowest supply of total water meeting estimate ET_0 (Table 3). It is also important to note that in 2022 at Logan, yields were not statistically different, yet Full irrigation was the least efficient irrigation dose. As well, at Cedar City, yield was

different in both seasons, however, WUE was not different among irrigation doses in either year.

Irrigation dose also affected alfalfa forage CP and RFV in nearly all site-years (Table 4; Fig. 5). In general, lower irrigation amounts were associated with higher CP and RFV, with the greatest values often observed in the Limited U treatment. These results are indicative of a physiological response of alfalfa growth, where alfalfa will allocate more resources to stem and vegetative structural growth than to leaves, reducing, then increasing CP and RFV values in a parabolic relationship as yield increases (Liu et al., 2021). However, under water deficit, there is a greater leaf to stem ratio, leading to increased RFV and CP when yields are low (Halim et al., 1989). These patterns in alfalfa growth bring attention to a critical trade-off between potential income from increased yield and greater quality and resource efficiency.

3.6. Strategies for optimal alfalfa production

The forgoing discussion largely presented yield and quality outcomes independently. However, to capture the balance of climate resilient strategies in irrigated agriculture that align efficiency and quality with sustainability (Hanson et al., 2007; Fereres and Soriano, 2007; Geerts and Raes, 2009; Yost et al., 2020), a water use efficiency metric adjusted by relative feed value was evaluated using decision tree models in this final section (WUE_{RFV} ; Fig. 6). The decision trees used the proportion of ET_0 met by irrigation (IrrET, %; Table 3) along with the irrigation technologies and alfalfa genetics to illustrate how different thresholds and factors impact the average WUE_{RFV} for each study site. Thus, the models provide practical “if-then” rules for stakeholders and growers who intend to meet estimated reference crop climatic demand (ET_0) with irrigation.

The Logan model had ten decision splits, while Cedar City and Vernal each had eight splits, indicating greater variation in the Logan subset, likely due to an additional growing season. The WUE_{RFV} at Logan during the study ranged from 0.22 to 3.1 and averaged 1.3. At Cedar City the WUE_{RFV} was between 0.13 and 0.92 with a mean of 0.5. However, at Vernal the range was 0.28–1.7 and a 0.93 mean. The model for Logan estimated that the highest average WUE_{RFV} was achieved when the proportion of IrrET was between 26 % and 32 % using LENA, LEPA, or MESA technologies. At Cedar City, the highest average WUE_{RFV} was 0.66 when the proportion of ET_0 met by irrigation was less than 49 %. Whereas, in Vernal, the highest average value-adjusted efficiency was estimated at 1.3 and predicted to be achieved when irrigation is applied to meet between 27 % and 32 % of crop climatic demand.

Among the model predictions for WUE_{RFV} values less than the highest average were reached through more nuanced decision rules that selected irrigation technology or alfalfa genetics, especially in Cedar City and Vernal. At Logan, the LENA, LEPA, or MESA technologies improved WUE_{RFV} values more than MDI and occasionally LESA. At Cedar City, where a greater proportion of crop climatic demand was met through irrigation than at Logan or Vernal, the conventional genetics, with less than 43 % IrrET had higher WUE_{RFV} than the DT genetics. When IrrET was greater than 53 % at Cedar City, LEPA and LESA reached higher WUE_{RFV} than LENA and MESA. At Vernal, when IrrET was between 39 % and 55 %, WUE_{RFV} values reached 0.95 group average for the LEPA, LESA, or MESA technologies. However, when IrrET was approximately 32 %, LENA or MDI improved WUE_{RFV} at Vernal.

Across the three models, maximizing feed value and efficiency principally depended on the amount of water applied to the crop in each environment, more so in Logan than in Cedar City or Vernal. For example, 80 % of the splits in the Logan model used IrrET rather than the irrigation technology or genetics factors; the Cedar City and Vernal models used IrrET at 50 % of the splits. Further, there was not a particular technology that improved WUE_{RFV} across all scenarios and points to balancing goals of increased yield with planning for water use efficiency and hopefully improved feed value of alfalfa forage. This

Table 5

Average yield loss and water savings from deficit irrigation relative to full irrigation dose. Irrigation dose comparisons correspond with alfalfa yield means separations by site-year (see Fig. 5).

Site	Year	Irrigation Dose Comparison	Yield Loss	Irrigation Savings from Deficit
			%	
Cedar City	2021	F, M vs. LU, LT	26	65
Cedar City	2022	F, M, LT vs. LU	45	48
Logan	2020	F vs. M, LU, LT	26	72
Logan	2021	F, M vs. LU, LT	22	66
Vernal	2021	F vs. LU, LT	54	56
Vernal	2022	F vs. M, LU, LT	14	71

Note: F, Full irrigation; M, Moderate; LU, Limited Uniform; LT, Limited targeted.

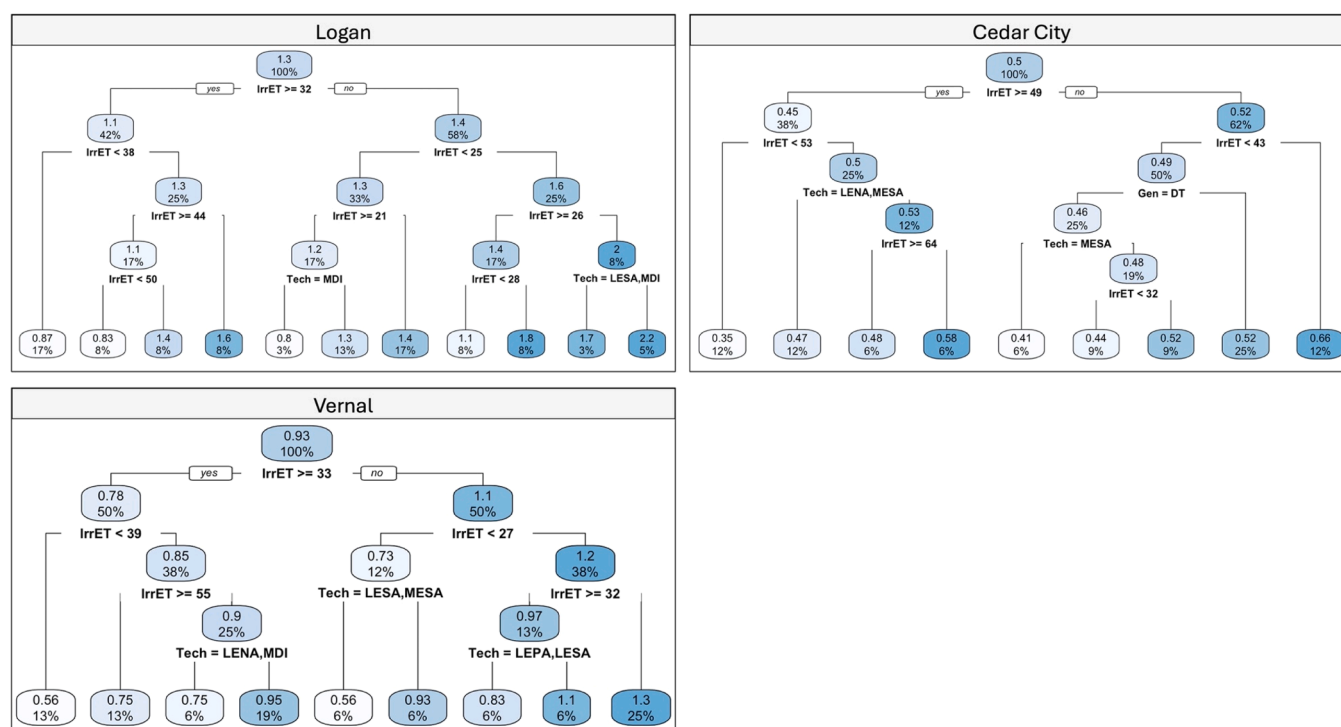


Fig. 6. Bootstrapped decision tree models for each site illustrating key thresholds for predictors of relative feed value-adjusted water use efficiency (WUE_{RFV}). Trees are read from the root node (top) to the leaf nodes (bottom). At each split, the site-specific dataset was divided based on a condition in the predictor variable (proportion of *ET*_o from irrigation, IrrET [%], irrigation technology, Tech, or alfalfa genetics, Gen). Nodes contain the predicted WUE_{RFV} and the proportion of observations in that node. Following any path from the root to a leaf node reveals how different levels and factors in the predictors impact the resulting average WUE_{RFV} for that decision tree branch.

study shows that the practical implications of choosing alfalfa production strategies are not simply a yield versus water trade-off and may be based more on alfalfa physiology than agronomic practices alone. The decision models showed the tendency of alfalfa protein, feed value, and water use efficiency to decrease with greater relative yield from increased applied irrigation as discussed above. Overall, the decision tree analysis demonstrated that choosing an irrigation technology to achieve optimal alfalfa production (defined by more water use efficient with higher feed value alfalfa) for the environment depends on first deciding the amount of estimated reference climatic demand that will be met by supplying irrigation.

4. Conclusions

This study examined the independent and combined effects of alfalfa genetics, irrigation technology, and irrigation dose on yield, forage quality, and water use efficiency across seven site-years in Utah. No significant interactions were detected among these factors, indicating that no cumulative or “stacked” benefits emerged from combining multiple water-optimizing practices in one- to three-year-old alfalfa stands. Instead, each practice influenced alfalfa performance independently, with varying effects across environments. Alfalfa drought-tolerant genetics did not consistently provide a yield advantage over conventional varieties in all environments. While the drought-tolerant Ladak II exhibited slightly improved water use efficiency in some site-years, yield was often lower during establishment years, suggesting that root development may be prioritized over vegetative production. Irrigation technology showed variable effects on yield and efficiency, with low-elevation systems (LEPA, LESA, and LENA) generally outperforming MESA and MDI. However, the advantage of these technologies was not universal across site-years. The decision tree models emphasized that alfalfa physiological responses depended largely on the proportion of *ET*_o met by irrigation rather than the specific technology

used, reinforcing the importance of holistic perspectives in site-specific irrigation management.

Deficit irrigation at a 25 % reduction maintain yields comparable to Full irrigation rate in half of the site-years, suggesting that a moderate deficit could be a viable water-saving strategy. However, reducing irrigation further (50 % reductions) consistently decreased yield, with no clear benefit in targeting irrigation at the beginning of the season (Limited T treatment) over a uniform application throughout the season (Limited U). Notably, deficit irrigation improved forage quality metrics, as lower irrigation doses were associated with higher CP and RFV, underscoring the trade-off between maximizing yield and optimizing quality. Additionally, the decision tree analysis revealed that achieving the highest relative feed value-adjusted water use efficiency depended primarily on matching irrigation to site-specific climatic demand rather than relying solely on genetics or irrigation delivery technology. In Logan, the highest WUE_{RFV} was observed when irrigation met 26–32 % of *ET*_o, while at Cedar City and Vernal, highest efficiency was reached at slightly higher irrigation thresholds. These findings highlight that irrigation strategies should be tailored to local environment conditions.

Overall, this study provided insights into the multifaceted challenges of optimizing alfalfa production under water-limited conditions. While no single factor pointed toward a universally applicable optimization tool, growers using drought-tolerant alfalfa genetics can anticipate lower yields until the stand is established, and low-elevation irrigation technologies can likely improve yield in years of normal precipitation once the stand is established. Additionally, a moderate irrigation deficit may offer promising water savings with minimal yield impacts. Future research should explore long-term on established stands and further refine strategies for optimizing alfalfa resilience and sustainability.

CRedit authorship contribution statement

Boren Dakota: Writing – original draft, Investigation, Data curation.

Yost Matt: Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Sullivan Tina:** Writing – review & editing, Investigation, Data curation. **Crookston Bradley:** Writing – review & editing, Visualization, Original draft, Data curation, Formal analysis. **Creech Earl:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Barker Burdette:** Writing – review & editing, Supervision. **Reid Cheyenne:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no competing interests related to the publication of this manuscript.

Acknowledgements

The authors express sincere gratitude for the many farm managers, cooperating farmers, and students who assisted with this research. This research was supported by USDA-Western Sustainable Agriculture and Research, Foundation for Food and Agriculture Research Young Innovator Award, Central Utah Water Conservancy District, USDA-Natural Resource Conservation Service, Utah State University Extension, and was supported and approved by the Utah Agricultural Experiment Station as journal paper number #9701.

Data availability

Data will be made available on request.

References

- Agweather, Utah, 2022. Utah Clim. Cent. (<https://climate.usu.edu/index.php>).
- Amosson, S., Almas, L., Girase, J.R., Vimlesh, K., Marek, T., 2011. Economics of irrigation systems. *Tex. A&M Agric. Commun. Ext. Publ. Agric. Org.* (<https://cdn.agrilifetoday.tamu.edu/wp-content/uploads/2011/10/Irrigation-Bulletin-FINAL-B6113.pdf>).
- Anower, M.R., Boe, A., Auger, D., Mott, I.W., Peel, M.D., Xu, L., Kanchupati, P., Wu, Y., 2017. Comparative drought response in eleven diverse alfalfa accessions. *J. Agron. Crop Sci.* 203 (1), 1–13. <https://doi.org/10.1111/jac.12156>.
- Breiman, L., 2017. Classification and regression trees. Routledge.
- Cardon, G., Jotuby-Amacher, J., Hole, P., Koenig, R., 2008. Understanding your soil test report. Utah State University Extension. AG/Soils/2008-01pr.
- Colby, B., Jones, L., O'Donnell, M., 2014. Supply reliability under climate change: Forbearance agreements and measurement of water conserved. In: Easter, K., Huang, Q. (Eds.), *Water markets for the 21st Century*. Global issues in water policy, v. 11. Springer, Dordrecht. https://doi.org/10.1007/978-94-017-9081-9_4.
- Dieter, C.A., Maupin, M.A., Caldwell, R.R., Harris, M.A., Ivahnenko, T.I., Lovelace, J.K., Barber, N.L., Linsey, K.S., 2017. Estimated use of water in the United States in 2015 (Circular 1441). U.S. Geological Survey. <https://doi.org/10.3133/cir1441>.
- Dietz, K.J., Zörb, C., Geilfus, C.M., 2021. Drought and crop yield. *Plant Biol.* 23 (6), 881–893. <https://doi.org/10.1111/plb.13304>.
- Fereres, E., Soriano, M.A., 2007. Deficit irrigation for reducing agricultural water use. *J. Exp. Bot.* 58 (2), 147–159. <https://doi.org/10.1093/jxb/erl165>.
- Frate, C.A., Roberts, B.A., Marble, V.L., 1991. Imposed drought stress has no long-term effect on established alfalfa. *Calif. Agric.* 45 (3), 33–36.
- Geerts, S., Raes, D., 2009. Deficit irrigation as an on-farm strategy to maximize crop water productivity in dry areas. *Agric. Water Manag.* 96 (9), 1275–1284. <https://doi.org/10.1016/j.agwat.2009.04.009>.
- Halim, R.A., Buxton, D.R., Hattendorf, M.J., Carlson, R.E., 1989. Water-stress effects on alfalfa forage quality after adjustment for maturity differences. *Agron. J.* 81, 189–194. <https://doi.org/10.2134/agronj1989.00021962008100020010x>.
- Hanson, B., Putnam, D., Snyder, R., 2007. Deficit irrigation of alfalfa as a strategy for providing water for water-short areas. *Agric. Water Manag.* 93 (1–2), 73–80. <https://doi.org/10.1016/j.agwat.2007.06.009>.
- Holt, J., Yost, M., Allen, N., Creech, E., Winward, D., Sullivan, T., 2021. Utah State Univ. Ext. Fact Sheet. Guide Irrig. Appl. Syst. Pivots Laterals. (<https://extension.usu.edu/crops/research/irrigation-pivots-laterals.pdf>).
- Hussain, S., Hussain, S., Qadir, T., Khaliq, A., Shraf, U., Parveen, A., Saqib, M., Rafiq, M., 2019. Drought stress in plants: an overview on implications, tolerance mechanisms and agronomic mitigation strategies. *Plant Sci. Today* 6, 389–402.
- Kang, Y., Han, Y., Torres-Jerez, I., Wang, M., Tang, Y., Monteros, M., Udvardi, M., 2011. System responses to long-term drought and re-watering of two contrasting alfalfa varieties. *Plant J.* 68 (5), 871–889. <https://doi.org/10.1111/j.1365-3113.2011.04738.x>.
- KRWG News and Partners., 2014, May 27. NMSU drought-tolerant alfalfa created for New Mexico. KRWG Public Media New Mexico State. (<https://www.krwg.org/region-al/2014-05-27/nmsu-drought-tolerant-alfalfa-created-for-new-mexico>).
- Li, M., Liu, Y., Yan, H., Sui, R., 2017. Effects of Irrigation Amount on Alfalfa Yield and Quality with a Center-Pivot System. *Nat. Resour. Environ. Syst. Commun. ASABE* 60 (5), 1633–1644.
- Lindenmayer, B.R., Hansen, N.C., Brummer, J., Pritchett, J.G., 2011. Deficit irrigation of alfalfa for water-savings in the great plains and intermountain west: a review and analysis of the literature. *Agron. J.* 103 (1), 45–50. <https://doi.org/10.2134/agronj2010.0224>.
- Liu, M., Wang, Z., Mu, L., Xu, R., Yang, H., 2021. Effect of regulated deficit irrigation on alfalfa performance under two irrigation systems in the inland arid area of midwestern China. *Agric. Water Manag.* 248. <https://doi.org/10.1016/j.agwat.2021.106764>.
- Milborrow, S., 2018. Package 'rpart.plot'. Plot 'rpart' Models: An Enhanced Version of 'plot.rpart'. Available online: (http://mercury.webster.edu/aleshunaz/R_learning_in_frastructure/code/rpart.plot.pdf).
- Montazar, A., Bachie, O., Corwin, D., Putnam, D., 2020. Feasibility of moderate deficit irrigation as a water conservation tool. *Agronomy* 10 (11), 1640. <https://doi.org/10.3390/agronomy10111640>.
- O'Geen, T., 2020. SoilWeb. University of California-Davis. Davis, CA. Accessed 23 Oct 2022. (<https://casoilresource.lawr.ucdavis.edu/gmap/>).
- O'Shaughnessy, S., Colaizzi, P., 2017. Performance of precision mobile drip irrigation in the Texas high plains region. *Agronomy* 7 (4). <https://doi.org/10.3390/agronomy7040068>.
- Oker, T.E., Kisekka, I., Sheshukov, A.Y., Aguilar, J., Rogers, D., 2020. Evaluation of dynamic uniformity and application efficiency of mobile drip irrigation. *Irrig. Sci.* 38 (1), 17–35. <https://doi.org/10.1007/s00271-019-00648-0>.
- Orloff, S., Putnam, D., Hanson, B., Carlson, H., 2005. California Alfalfa and Forage Symposium. Implic. Deficit Irrig. Manag. Alfalfa. (<http://alfalfa.ucdavis.edu>).
- Ottman, M.J., Putnam, D.H., 2017. Deficit irrigation with alfalfa: What are the economics. In *Proceedings for the 47th Western Alfalfa & Grains Symposium* (<http://alfalfa.ucdavis.edu>).
- Radawich, J.A., 2018. Examining the interactions of alfalfa varieties and deficit irrigation strategies under subsurface drip irrigation. University of California, Davis. ProQuest Dissertations Publishing, 10937885.
- Samani, Z., Skaggs, R.K., 2006. The multiple personalities of water conservation. *Water Policy* 10 (3), 265–294. <https://doi.org/10.2166/wp.2008.154>.
- Seager, R., Ting, M., Alexander, P., Nakamura, J., Liu, H., Li, C., Simpson, I.R., 2022. Mechanisms of a meteorological drought onset: summer 2020 to spring 2021 in Southwestern North America. *J. Clim.* 35, 7367–7385. <https://doi.org/10.1175/JCLI-D2-22-0314.1>.
- Sitterson, J.M., Andales, A.A., Mooney, D.F., Capurro, M.C., Brummer, J.E., 2023. Developing a crop water production function for alfalfa under deficit irrigation: a case study in Eastern Colorado. *Agriculture* 13 (4), 831. <https://doi.org/10.3390/agriculture13040831>.
- Sojka, R.E., Bjorneberg, D.L., Entry, J.A., 2002. Irrigation: an historical perspective. *Encycl. Soil Sci.* 745–749.
- Taghvaeian, S., Frazier, S., Livingston, D., Fox, G., 2017. Ogallala Aquifer. (<http://osu.facts.okstate.edu>).
- Therneau, T., Atkinson, B., Ripley, B., & Ripley, M.B., 2015. Package 'rpart'. Available online: (<http://cran.ma.ic.ac.uk/web/packages/rpart/rpart.pdf>).
- Tibshirani, R.J., Efron, B., 1993. An introduction to the bootstrap. *Monogr. Stat. Appl. Probab.* 57 (1), 1–436.
- Trout, T.J., Howell, T.A., English, M.J., Martin, D.L., 2020. Deficit irrigation strategies for the Western U.S. *Am. Soc. Agric. Biol. Eng.* 63 (6), 1813–1825.
- U.S.D.A. National Agricultural Statistic Service (USDA-NASS), 2021. Data and statistics: Quick stats. USDA-NASS. (<https://www.nass.usda.gov/QuickStats/>).
- Udall, B., Peterson, G., 2017. "Part 1 of 5. Executive Summary." in *Agricultural Water Conservation in the Colorado River Basin: Alternatives to Permanent Fallowing Research Synthesis and Outreach Workshops*. Colorado State University Colorado Water Institute, Fort Collins, CO. Available at: (http://www.crbagwater.colostate.edu/files/CWI_Completion_Report232_Part1.pdf).
- USDA Agricultural Marketing Service (USDA-AMS), 2023. Hay quality designation guidelines. Livestock, poultry, and grain market news. (<https://www.ams.usda.gov/sites/default/files/media/HayQualityGuidelines.pdf>).
- Waller, P., Yitayew, M., 2016. Irrigation and Drainage Engineering. Springer International Publishing Switzerland. <https://doi.org/10.1007/978-3-319-05699-9>.
- Ward, R.A., Salisbury, K., 2016. The Economic Contribution of Agriculture to the Utah Economy in 2014, Utah State University Economic Research Institute Report #2016-01, October 2016.
- Williams, A.P., Cook, B.I., Smerdon, J.E., 2022. Rapid intensification of the emerging southwestern North American megadrought in 2020–2021. *Nat. Clim. Chang.* 12, 232–234. <https://doi.org/10.1038/s41558-022-01290-z>.
- Yost, M., Allen, N., Peterson, W., Gale, J., 2020. Water conservation through applied irrigation research. 75 Year of conservation special edition. *J. Soil Water Conserv.* (https://www.swcs.org/static/media/cms/75th_Book_online_small_EB8351774A8B2.pdf).